Does Urban Status Reverse the Poverty-Income Effect on Low Food Access? A Modelling Investigation of the Food Access Research Atlas

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1 Executive Summary

[TODO: write this LAST, once the Detailed Analysis results exist. One paragraph: what question, what method, what was found, how confident.]

2 Background

[TODO: expand on the group report's Background, but focused on YOUR question. Suggested angle based on the group EDA: pooled, PovertyRate and MedianFamilyIncome barely separate LA1and10 (Cohen's d ~0.2-0.27) - but splitting by Urban reverses the direction of both relationships (Simpson's paradox, see group_report). State the specific research question this motivates, e.g.: "Does modelling LA1and10 with an Urban interaction term change which predictors matter, compared to a pooled model?"]

2.1 Research question(s)

[TODO: state the specific question(s) precisely, in a form you can actually test.]

3 Data Description

[TODO: revise from the group report's Background/Data section - state structure + completeness again briefly, then focus on the specific variables your Detailed Analysis actually uses (e.g. PovertyRate, MedianFamilyIncome, Urban, LA1and10) and what their categories mean.]

4 Ethics, Privacy and Security

[TODO: carry over from group_report.Rmd once feedback is received on the Phase 1 report; adjust wording if needed.]

5 Exploratory Data Analysis

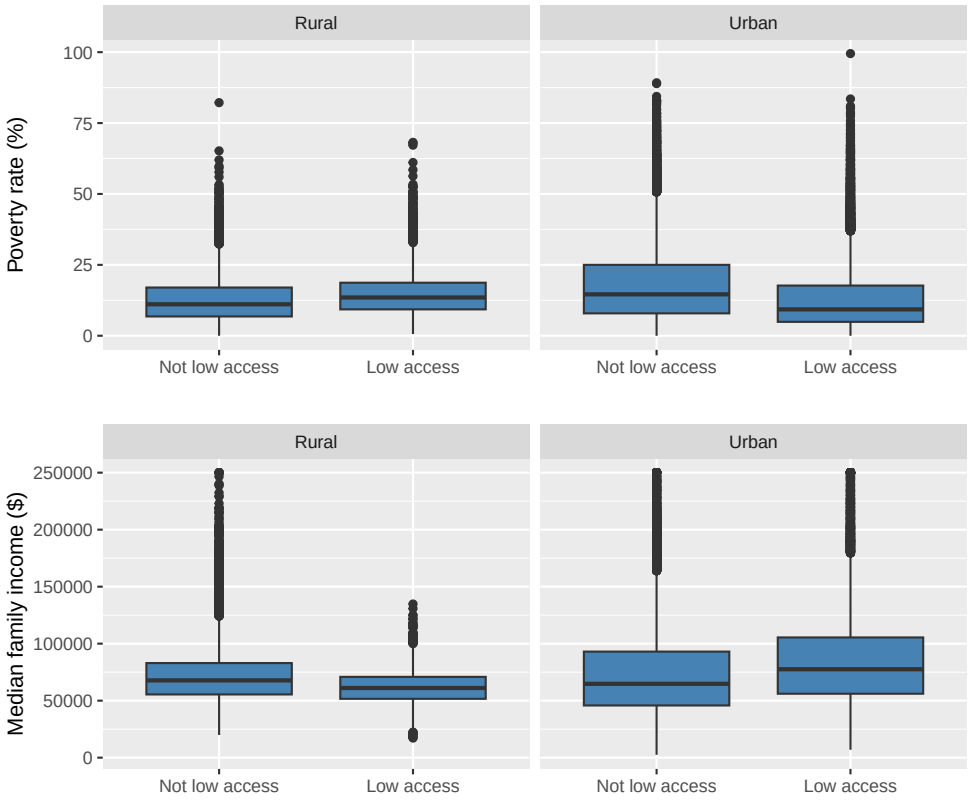


Figure 1: [TODO: write final caption] PovertyRate and MedianFamilyIncome by LA1and10 status, split by Urban/Rural – the pattern reverses direction between the two panels.

6 Detailed Analysis Results

6.1 Method

[TODO: what technique, why it fits the question. Candidate given the Simpson's-paradox motivation: logistic regression of LA1and10 on PovertyRate + MedianFamilyIncome + Urban + an Urban interaction term, compared against the same model without the interaction (likelihood-ratio test on the interaction term = the formal hypothesis test).]

6.2 Data preparation for this analysis

[TODO: any transformations - e.g. standardising PovertyRate/Income before including them in a model, log-transform if skew is an issue, how missing data was handled here specifically.]

6.3 Model specification

[TODO: full list of terms in the model, exactly as fitted - e.g. `LA1and10 ~ PovertyRate * Urban + MedianFamilyIncome * Urban`.]

Table 1: Variance inflation factors, M0 (main effects)

Term	VIF
PovertyRate_z	1.83
MedianFamilyIncome_z	1.83
Urban_f	1.05

Table 2: M1 (with Urban interaction): odds ratios and 95% CIs

Term	Odds Ratio	Lower CI	Upper CI	p-value
(Intercept)	0.156	0.146	0.166	<0.001
PovertyRate_z	0.787	0.727	0.852	<0.001
Urban_fUrban	5.127	4.809	5.474	<0.001
MedianFamilyIncome_z	0.303	0.267	0.343	<0.001
PovertyRate_z:Urban_fUrban	0.812	0.748	0.883	<0.001
Urban_fUrban:MedianFamilyIncome_z	3.141	2.771	3.568	<0.001

Table 3: Model fit comparison

Model	AIC	McFadden pseudo-R2
M0: main effects only	89551.9	0.0608
M1: with Urban interaction	88397.0	0.0730

Table 4: Likelihood-ratio test: M1 vs M0

Resid. Df	Deviance	Pr(>Chi)
71779	NA	NA
71777	1158.91	<0.001

6.4 Results (M0 vs M1)

[TODO: write this once you've checked the tables above render sensibly. Suggested structure: (1) state the LRT p-value and what it means for the "does the relationship differ by Urban/Rural" question, (2) walk through M1's odds ratios for PovertyRate_z, MedianFamilyIncome_z, and their Urban interaction terms – note that for a factor interaction, the Urban-specific slope is the main-effect coefficient PLUS the interaction coefficient, not the interaction coefficient alone, (3) compare AIC/ pseudo-R2 between the two models to justify keeping the interaction, (4) note VIF result and whether MedianFamilyIncome was kept or dropped.]

6.5 M2: does vehicle access explain the reversal?

[TODO: state the mediation-style question – does adding vehicle access attenuate the Urban interaction coefficients, i.e. is limited vehicle access part of *why* the relationship flips by Urban rather than Urban being a standalone confound?]

Table 5: Sample size used for the M2 comparison (structural missingness in lahunv1share)

Dataset	Rows	% of full sample
Full sample (M0/M1)	71783	100.0
lahunv1share complete cases (M1 refit / M2)	52047	72.5

Table 6: Urban interaction coefficients before/after adding lahunv1share

Model	Term	Estimate	Std. Error	p-value
M1 (refit on M2 subsample)	Poverty x Urban	0.013	0.043	0.762
M1 (refit on M2 subsample)	Income x Urban	1.261	0.065	<0.001
M2 (+ lahunv1share)	Poverty x Urban	0.534	0.051	<0.001
M2 (+ lahunv1share)	Income x Urban	1.239	0.070	<0.001

[TODO: read off the table above – did the interaction coefficients shrink toward 0 from M1 (refit) to M2? By roughly how much (%)? State the conclusion for the mediation-style question.]

6.6 Goodness of fit (chosen model: M1)

Table 7: Hosmer-Lemeshow goodness-of-fit test for M1

Statistic	value	df	p-value
Hosmer-Lemeshow X-squared	634.18	8	<0.001

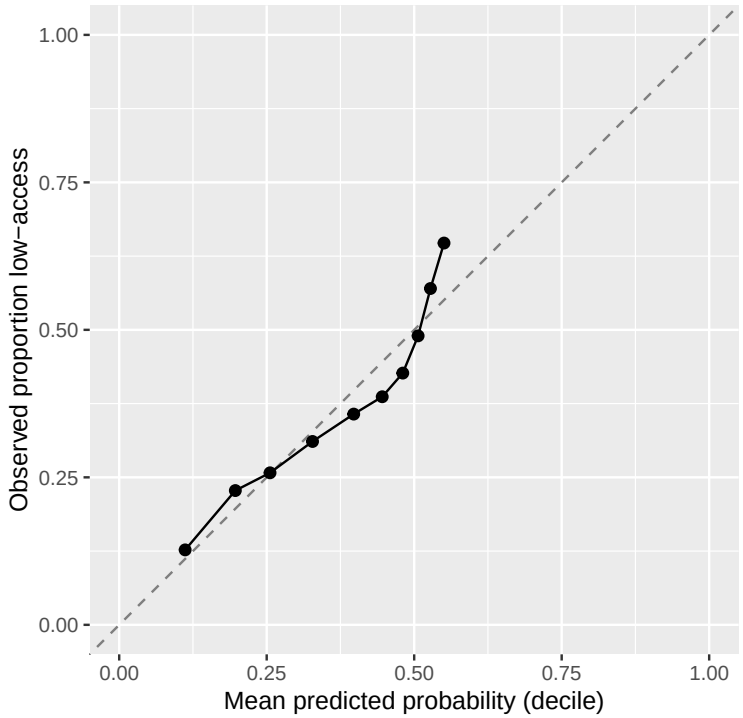


Figure 2: Calibration plot for M1: predicted vs. observed low-access rate by decile of predicted probability. Points on the dashed line indicate perfect calibration.

[TODO: state the HL test p-value and what it means (large p = fail to reject H0 of good calibration – but note that at very large n, HL becomes extremely sensitive and will flag even tiny, practically irrelevant deviations from perfect calibration as "significant", so a significant HL result here doesn't necessarily mean the model is practically miscalibrated – read the calibration plot alongside it for the practical picture, not instead of it. With n ~ 72,000 here, expect HL to reject even for a genuinely well-calibrated model – that's why the calibration plot matters more than the p-value at this sample size).]

6.7 M3: random forest comparison, with GA-based feature selection

[TODO: explain why a second, non-parametric method is worth fitting – to check whether the logistic model's interaction-term story survives in a method that doesn't assume a particular functional form, and to satisfy the brief's "ideally fit a prediction/ML model" ask with an actual second method, not a third logistic-regression variant.]

6.7.1 Feature selection: genetic algorithm

[TODO: explain the GA – following the approach from AIML426 (Evolutionary Computation), each chromosome is a binary vector marking which optional predictors are included; PovertyRate_z, MedianFamilyIncome_z and Urban are always included since they're central to the research question, and the GA searches over whether to add each of the four demographic composition shares. Fitness = 5-fold cross-validated AUC of a random forest fit on the selected features.]

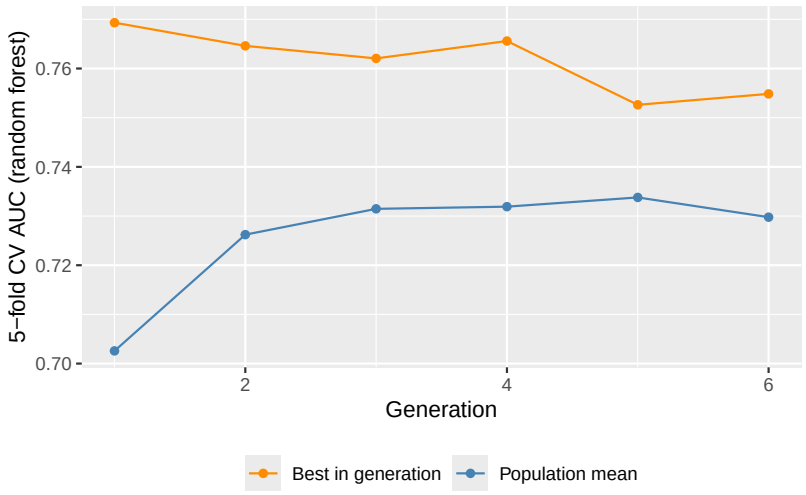


Figure 3: Genetic algorithm convergence: best and mean 5-fold CV AUC by generation.

Table 8: Features selected by the genetic algorithm

Feature	Type	Selected
PovertyRate_z	Mandatory	Yes
MedianFamilyIncome_z	Mandatory	Yes
Urban_f	Mandatory	Yes
black_share_plain_z	Optional (GA-selected)	Yes
kids_share_plain_z	Optional (GA-selected)	Yes
seniors_share_plain_z	Optional (GA-selected)	Yes
hunv_share_plain_z	Optional (GA-selected)	Yes

[TODO: state which optional features the GA kept, and whether that's substantively interesting (e.g. did it keep the same demographic shares that mattered in the group EDA's correlation heatmap?).]

6.7.2 Final random forest vs. M1: held-out test performance

Table 9: Held-out test performance: M1 vs. M3 (bootstrap 95% CI on AUC)

Model	AUC	95% CI	Test Accuracy	Test n
M1 (logistic)	0.686	(0.679, 0.693)	0.662	21536
M3 (random forest)	0.776	(0.769, 0.782)	0.711	21536

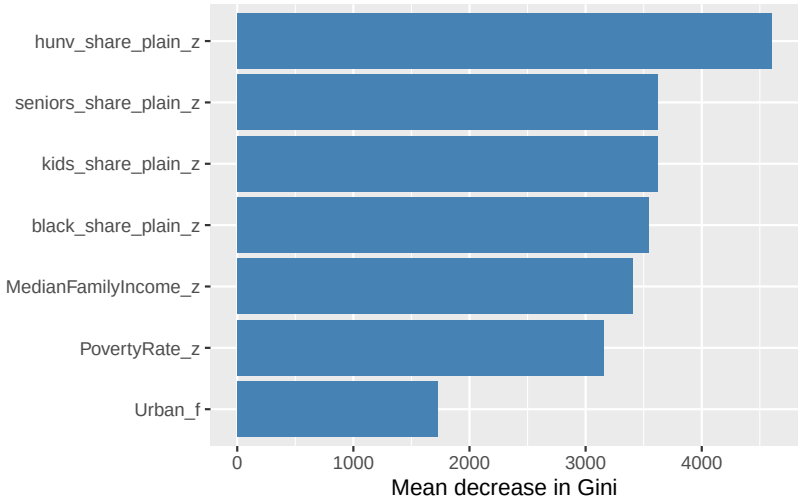


Figure 4: Random forest variable importance (mean decrease in Gini).

6.8 Results (M3)

[TODO: (1) does the random forest's test AUC overlap M1's CI, or is one clearly better – what does that say about whether the logistic interaction structure captures the relationship well? (2) does the GA-selected feature importance ranking agree with which predictors mattered in M1 and the group EDA's correlation heatmap? (3) note that random forest importance and logistic-regression coefficients answer different questions (predictive contribution vs. a controlled partial effect) – don't conflate them. (4) biases: tract-level aggregation (ecological fallacy risk), MedianFamilyIncome top-coded at $250,001, M2's non-random missingness (noted above).]

7 Conclusions and Recommendations

[TODO: main conclusion(s), any recommendation, limitations, optionally future work.]

8 References

[TODO: at minimum, cite the data source, e.g.

United States Department of Agriculture, Economic Research Service (2019). *Food Access Research Atlas*. https://www.ers.usda.gov/data-products/food-access-research-atlas/]